

D300

Identifying Learner Needs and a Research Problem

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A designer at a community college plans a study of an online math refresher. She will pilot it with adult learners in real courses, revise it after each term based on data, and write up principles other designers can reuse. The college recently renovated its library and has a large parking garage. Which feature makes this design-based research?

- A. Building and launching the refresher with no data collection.
- B. Repeated revise-and-test cycles in real courses that produce both a better solution and reusable principles.
- C. A single test in a controlled lab with random assignment.
- D. Studying how the library renovation affects learning.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A researcher tests a finished tutorial once, reports the score change, and ends the project. What is missing for this to count as design-based research?

- A. Cycles of revision based on the data, and principles drawn from them.
- B. A control group with random assignment.
- C. A larger sample size.
- D. Nothing; any study of a tutorial is design-based research.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A training department wants 90% of new support agents to resolve tickets on first contact. Agents currently resolve 62% on first contact. The department has a new manager and a modern open-plan office. How should the learning need be stated?

- A. A 90% gap in first-contact resolution.
- B. Agents need a new video training series.
- C. A 28-percentage-point gap in first-contact resolution, from 62% to 90%.
- D. Agents need to adjust to the open-plan office.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Current on-time report submission is 55%, and the gap to the goal is 30 percentage points. What is the desired performance level?

- A.** 85%
- B.** 100%, because goals are always full compliance.
- C.** 25%
- D.** 30%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A designer is starting the empathize phase for an e-learning module aimed at adult learners returning to school. The program director, who has led the program for ten years, says she already knows learners just need more videos. The program's website was redesigned last spring. What should the designer do?

- A.** Start building the videos to save time.
- B.** Review the redesigned website's analytics only.
- C.** Accept the director's view, given her ten years of experience.
- D.** Interview and observe several returning adult learners about their experience and struggles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

After interviews, a designer learns that part-time adult learners at a large online program often miss key announcements because they log in only on weekends. The program serves learners in 30 countries. Which is the best define-stage problem statement?

- A.** Learners in 30 countries need translated announcements.
- B.** Learners need to log in more.
- C.** We need to build a mobile app with push notifications.
- D.** Part-time adult learners who log in only on weekends need a way to catch time-sensitive information because announcements are posted midweek.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team's draft problem statement reads: "We need to create a gamified leaderboard for new hires." What is the main flaw?

- A.** It is too short; problem statements must be at least a paragraph.
- B.** It should be written as a research hypothesis.
- C.** It states a solution and says nothing about the learners' need or the insight behind it.
- D.** It should name a specific gaming platform.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A designer has defined the problem: new remote employees at a mid-size firm struggle to apply a data-security policy. The firm uses three video platforms and has an award-winning intranet. Which research question is best?

- A.** What is the best way to fix all security problems in every company?
- B.** How do new remote employees describe the barriers to applying the data-security policy in their daily work?
- C.** Is security important?
- D.** Which of the three video platforms has the best features?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A student's research question is "Is online learning good?" Which revision most improves it?

- A.** Is face-to-face learning good?
- B.** How does adding weekly scenario practice affect first-year adult learners' scores on the program's writing rubric?
- C.** Is online learning very good?
- D.** Why is online learning better than all other formats?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A designer needs evidence that spaced practice helps adult learners retain procedures. She finds four sources, one with a colorful infographic and one written by a well-known consultant. Which source best supports the claim in a literature review?

- A.** A consultant’s blog post endorsing spaced practice.
- B.** A vendor white paper with an infographic about its spaced-practice tool.
- C.** A textbook glossary definition of spaced practice.
- D.** A peer-reviewed journal study that measured retention in adults who used spaced practice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A literature review finds many peer-reviewed studies showing that microlearning helps adults recall facts, but almost none on whether it helps adults transfer complex procedures to the job. The reviewer read 48 articles over two months. What should the reviewer conclude?

- A.** The gap is microlearning’s effect on transfer of complex procedures; that is where a new study adds value.
- B.** Repeat the fact-recall studies with a new group.
- C.** Microlearning is well researched, so no further study is needed.
- D.** Read 48 more articles before deciding anything.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A draft literature review has twelve paragraphs; each summarizes one article, starting with the author and year. The writer used a reference manager and a style guide. What revision most improves the review?

- A.** Add more articles, one paragraph each.
- B.** Sort the paragraphs by publication year.
- C.** Reorganize by themes, comparing what several studies found on each theme.
- D.** Switch to a different reference manager.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A designer at a community college plans a study of an online math refresher. She will pilot it with adult learners in real courses, revise it after each term based on data, and write up principles other designers can reuse. ~~The college recently renovated its library and has a large parking garage.~~ Which feature makes this design-based research?

Target word: defining feature of DBR **Decides:** Iterative cycles in a real setting that yield a solution and design principles.

A	Dropped element	Leaves out the research half of DBR.
B	Correct	That dual goal in an authentic, iterative setting defines DBR.
C	Adjacent category	Describes a classic experiment, not DBR.
D	Camouflage	Pulls in an irrelevant campus detail.

2. Answer A Concept 1 • transfer

A researcher tests a finished tutorial once, reports the score change, and ends the project. What is missing for this to count as design-based research?

Target word: reverse: what is absent **Decides:** No iteration and no design principles.

A	Correct	DBR requires iteration and theory-building.
B	Confusion set	That strengthens an experiment; it is not DBR's defining feature.
C	Adjacent category	Size does not make a study design-based.
D	Absolute language	Studying a product once is not DBR.

3. Answer C Concept 2

A training department wants 90% of new support agents to resolve tickets on first contact. Agents currently resolve 62% on first contact. ~~The department has a new manager and a modern open-plan office.~~ How should the learning need be stated?

Target word: state the gap **Decides:** Desired 90% vs. current 62%.

A	Wrong stem number	Uses the goal as the gap.
B	Inverted operation	States a solution, not a need.
C	Correct	$90 - 62 = 28$ points; stated as a measurable gap.
D	Camouflage	The office layout is not the measured gap.

4. Answer A Concept 2 • transfer

Current on-time report submission is 55%, and the gap to the goal is 30 percentage points. What is the desired performance level?

Target word: solve for desired level **Decides:** Desired = current + gap.

A	Correct	$55 + 30 = 85$.
B	Absolute language	The stated gap fixes the goal at 85%.
C	Inverted operation	Subtracts the gap from current performance.
D	Wrong stem number	That is the gap itself.

5. Answer D Concept 3

A designer is starting the empathize phase for an e-learning module aimed at adult learners returning to school. The program director, ~~who has led the program for ten years~~, says she already knows learners just need more videos. ~~The program's website was redesigned last spring~~. What should the designer do?

Target word: best empathize step **Decides:** Empathy must come from the learners themselves.

A	Inverted operation	Jumps to a solution before understanding the need.
B	Adjacent category	Secondary data can help but does not replace empathizing with learners.
C	Camouflage	Experience does not replace hearing from learners.
D	Correct	Direct contact with learners grounds the problem in real needs.

6. Answer D Concept 4

After interviews, a designer learns that part-time adult learners at a large online program often miss key announcements because they log in only on weekends. ~~The program serves learners in 30 countries~~. Which is the best define-stage problem statement?

Target word: human-centered problem statement **Decides:** Names learner, need, and insight with no built-in solution.

A	Camouflage	Built on a detail the interviews did not identify as the issue.
B	Dropped element	Drops the insight and blames the learner.
C	Inverted operation	A solution, not a problem statement.
D	Correct	Learner, need, and insight; open to many solutions.

7. Answer C Concept 4 • transfer

A team's draft problem statement reads: "We need to create a gamified leaderboard for new hires." What is the main flaw?

Target word: reverse: diagnose a flawed statement **Decides:** A solution is written in place of a need.

A	Absolute language	Length is not the standard.
B	Adjacent category	A hypothesis is a different research element.
C	Correct	A define statement names learner, need, and insight, leaving solutions open.
D	Inverted operation	Makes the solution even more specific instead of removing it.

8. Answer B Concept 5

A designer has defined the problem: new remote employees at a mid-size firm struggle to apply a data-security policy. ~~The firm uses three video platforms and has an award-winning intranet~~. Which research question is best?

Target word: best research question **Decides:** Focused, answerable with collectible data, tied to the problem.

A	Absolute language	Far too broad to answer.
B	Correct	Focused on the defined problem and answerable through interviews or surveys.
C	Dropped element	A yes/no value question not tied to learners or data.
D	Camouflage	Chases an irrelevant detail, not the learning problem.

9. Answer B Concept 5 • transfer

A student's research question is "Is online learning good?" Which revision most improves it?

Target word: fix a weak question **Decides:** Needs focus, a population, and measurable terms.

A	Inverted operation	Flips the topic without fixing the flaws.
B	Correct	Specific population, intervention, and measure.
C	Dropped element	Still broad, yes/no, and value-based.
D	Absolute language	Assumes the answer and overgeneralizes.

10. Answer D Concept 6

A designer needs evidence that spaced practice helps adult learners retain procedures. She finds four sources, ~~one with a colorful infographic and one written by a well-known consultant~~. Which source best supports the claim in a literature review?

Target word: strongest source type **Decides:** A peer-reviewed study that collected data on adult learners.

A	Camouflage	Author fame does not make an opinion empirical.
B	Adjacent category	Marketing material, not peer-reviewed research.
C	Confusion set	Defines the term but gives no evidence that it works.
D	Correct	Empirical, peer-reviewed evidence on the claim.

11. Answer A Concept 7

A literature review finds many peer-reviewed studies showing that microlearning helps adults recall facts, but almost none on whether it helps adults transfer complex procedures to the job. ~~The reviewer read 48 articles over two months.~~ What should the reviewer conclude?

Target word: identify the gap **Decides:** Transfer of complex procedures is under-studied.

A	Correct	The unknown area justifies the research.
B	Inverted operation	Studies what is already known instead of the gap.
C	Dropped element	Ignores the unstudied transfer question.
D	Camouflage	The count of articles does not decide the gap.

12. Answer C Concept 8

A draft literature review has twelve paragraphs; each summarizes one article, starting with the author and year. ~~The writer used a reference manager and a style guide.~~ What revision most improves the review?

Target word: fix the organization **Decides:** Source-by-source summaries instead of synthesis.

A	Inverted operation	Makes the list longer without synthesizing.
B	Adjacent category	Still an annotated list, just reordered.
C	Correct	Synthesis connects sources around ideas.
D	Camouflage	The tool is not the problem.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	2	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	What makes research design-based	Design-based research tests and refines a real intervention in a real setting through repeated cycles, producing both a working solution and design principles.	Thinking it is a one-time lab experiment or just building a product.	Iteration in an authentic setting with a theory-building goal.
2	The performance gap	A learning need is the gap between desired and current performance, stated in measurable terms.	Stating a need as a solution (“learners need a video”).	A desired level and a current level of the same measure.
3	Empathize with actual learners	Empathy comes from observing and interviewing the real learners, not from designer assumptions or sponsor opinions.	Relying on what managers or experts say learners need.	A choice between talking to learners and talking about them.
4	A human-centered problem statement	The define-stage statement names the learner, their need, and the insight, without building in a solution.	Writing a solution into the problem statement.	“We need to build...” inside a problem statement.
5	A researchable question	A good research question is focused, answerable with data you can collect, and tied to the defined problem.	Asking broad, yes/no, or value questions that data cannot answer.	Choosing the best research question.
6	Empirical, peer-reviewed sources	A literature review rests on peer-reviewed empirical studies that collected and analyzed data; opinion pieces and vendor claims are not evidence.	Counting popular articles or vendor white papers as research.	Deciding which source can support a claim.
7	The gap in the literature	A review shows what is known and what is not; the unknown part justifies the study.	Thinking a topic with lots of research needs no new study, or one with none is automatically worth studying.	“What has already been done and what is unknown?”
8	Synthesis by theme	A literature review is organized by themes across sources, comparing findings, not one summary per article.	Writing an annotated list, source by source.	Paragraphs that each begin with a different author.

Course scope (public description): Identifying Learner Needs and a Research Problem is the first of three capstone courses in the program. This course provides an introduction to design-based research and focuses specifically on the first two phases of the design-based research process: identifying and analyzing the learning problem and reviewing the literature. This course also requires that learners continue applying Design Thinking as they empathize with learners and define the instructional problem that their research will help them understand and address. Finally, this course teaches learners how to conduct a literature review to determine what research has already been done and what is unknown about their research topic. Learning Experience Design Lab is a prerequisite for this course.